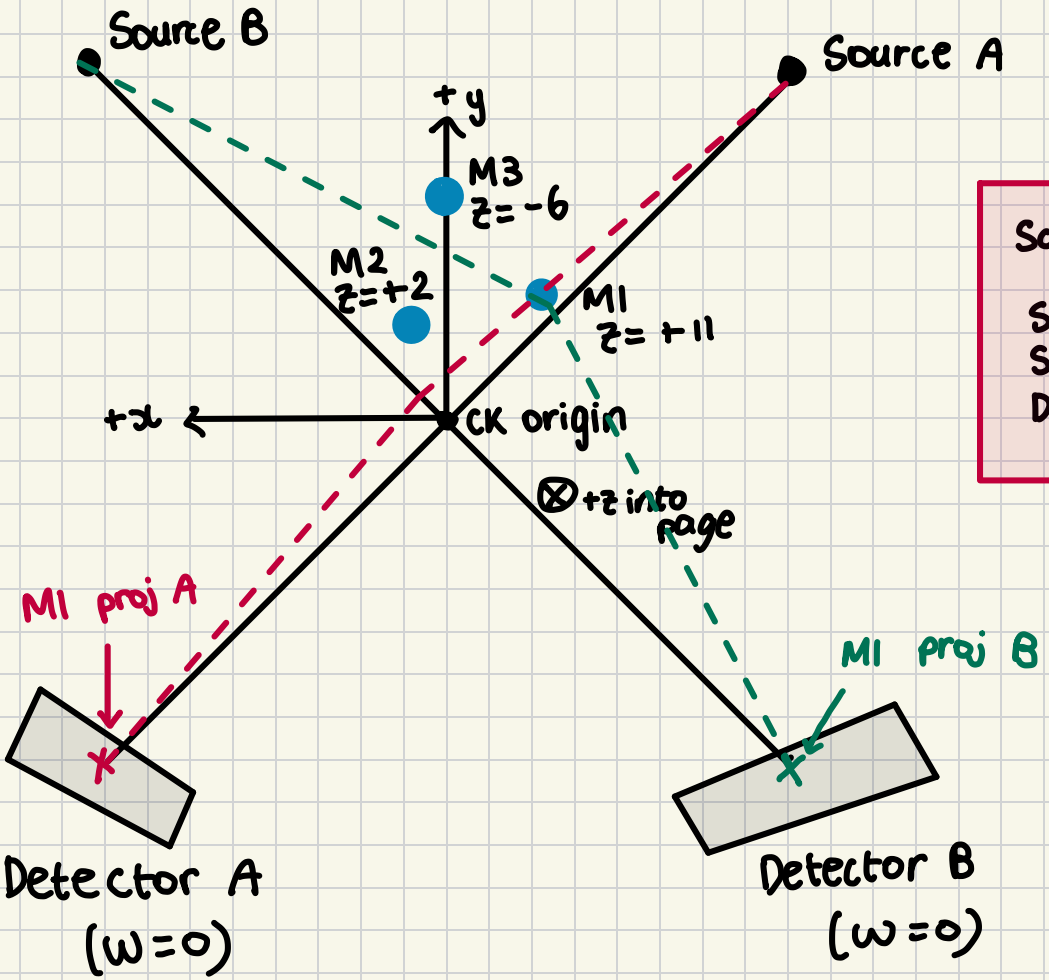


Schematic - not to scale

SAD = 1000 mm
SDD = 2000 mm
Detector plane: w=0

Hand Calculations

$$M = \frac{SDD}{SDD - w}$$
$$u_{proj} = Mu$$
$$v_{proj} = Mv$$

w/ SDD = 2000mm

for detector A: $u_A = z$, $v_A = \frac{x+y}{\sqrt{2}}$, $w_A = 1000 + \frac{y-x}{\sqrt{2}}$

for detector B: $u_B = z$, $v_B = \frac{x-y}{\sqrt{2}}$, $w_B = 1000 + \frac{x+y}{\sqrt{2}}$

M1 = (-19, 28, 11) mm

Detector A: $u_A = 11$

$$v_A = \frac{-19+28}{\sqrt{2}} = \frac{9}{\sqrt{2}} \approx 6.36$$
$$w_A = 1000 + \frac{28-(-19)}{\sqrt{2}} \approx 1033.23$$
$$M_A = \frac{2000}{2000-1033.23} \approx 2.069$$
$$u_{proj} = 11(2.069) \approx 22.76$$
$$v_{proj} = 6.36(2.069) \approx 13.17$$

$M1_A \approx (23, 13)$ mm

Detector B: $u_B = 11$

$$v_B = \frac{-19-28}{\sqrt{2}} \approx -33.23$$
$$w_B = 1000 + \frac{-19+28}{\sqrt{2}} \approx 1006.36$$
$$M_B = \frac{2000}{2000-1006.36} \approx 2.013$$
$$u_{proj} = 11(2.013) \approx 22.14$$
$$v_{proj} = -33.23(2.013) \approx -66.89$$

$M1_B \approx (22, -67)$ mm

M2 = (43, 14, 2) mm

Detector A: $u_A = 2$

$$v_A = \frac{43+14}{\sqrt{2}} \approx 40.31$$
$$w_A = 1000 + \frac{14-43}{\sqrt{2}} \approx 979.49$$
$$M_A = \frac{2000}{2000-979.49} \approx 1.960$$
$$u_{proj} = 2(1.960) \approx 3.92$$
$$v_{proj} = 40.31(1.960) \approx 78.99$$

$M2_A \approx (4, 79)$ mm

Detector B: $u_B = 2$

$$V_B = \frac{43-14}{\sqrt{2}} \approx 20.51$$

$$W_B = 1000 + \frac{43+14}{\sqrt{2}} \approx 1040.31$$

$$M_B = \frac{2000}{2000-1040.31} \approx 2.084$$

$$u_{proj} = 2(2.084) \approx 4.17$$

$$V_{proj} = 20.51(2.084) \approx 42.73$$

$$M2_B \approx (4, 43)\,mm$$

∴ computed projections closely match the hand estimates in sign, relative positive, and magnitude, confirming consistency.

$M3 = (2, 48, -6)\,mm$

Detector A: $u_A = -6$

$$V_A = \frac{2+48}{\sqrt{2}} \approx 35.36$$

$$W_A = 1000 + \frac{48-2}{\sqrt{2}} \approx 1032.52$$

$$M_A = \frac{2000}{2000-1032.53} \approx 2.067$$

$$u_{proj} = -6(2.067) \approx -12.40$$

$$V_{proj} = 35.36(2.067) \approx 73.09$$

$$M3_A \approx (-12, 73)\,mm$$

Detector B: $u_B = -6$

$$V_B = \frac{2-48}{\sqrt{2}} \approx -32.53$$

$$W_B = 1000 + \frac{2+48}{\sqrt{2}} \approx 1035.36$$

$$M_B = \frac{2000}{2000-1035.36} \approx 2.073$$

$$u_{proj} = -6(2.073) \approx -12.44$$

$$V_{proj} = -32.53(2.073) \approx -67.44$$

$$M3_B \approx (-12, -67)\,mm$$